



K-space is filled line by line using phase encoding and frequency encoding.

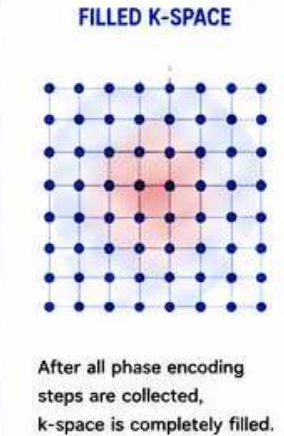
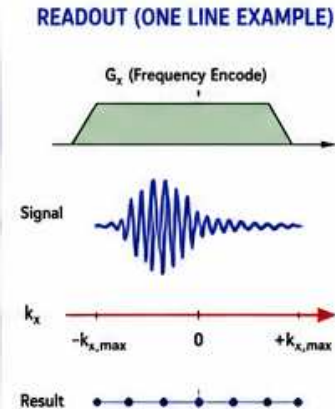
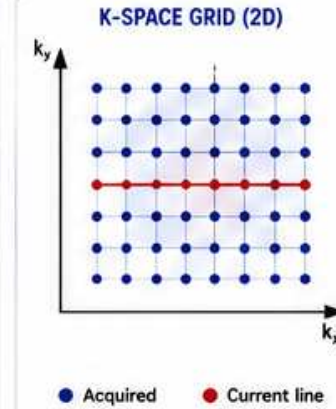
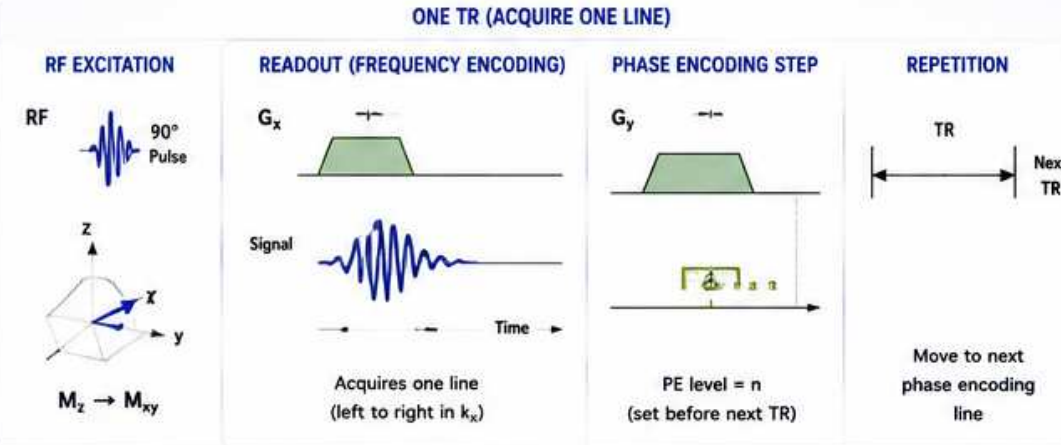


THE BIG IDEA

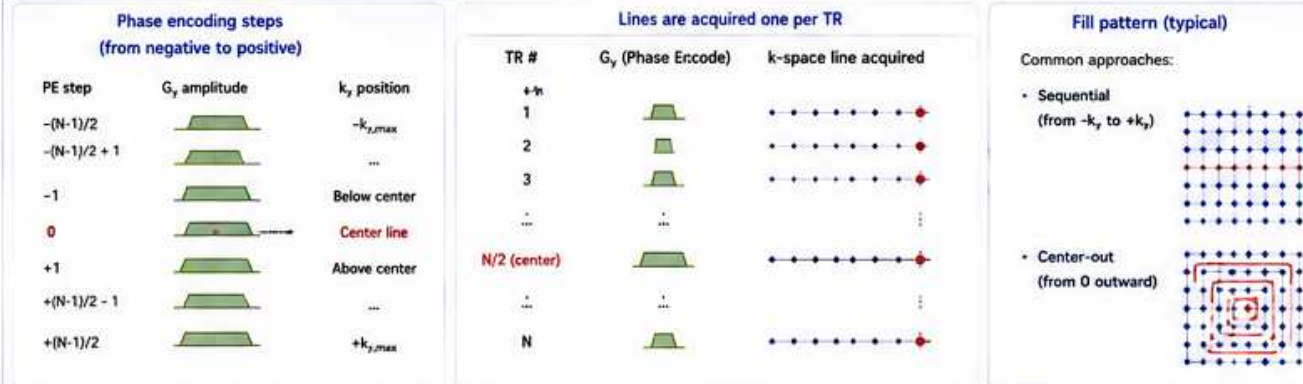
The readout (frequency encoding) gradient fills one line of k-space. The phase encoding gradient steps to a new value for the next line. After many TRs, k-space is completely filled.

THE CONCEPT

- 1 RF excitation tips M_z into the transverse plane.
- 2 While the signal is acquired, G_x (readout) is turned on to sample one line of k-space.
- 3 G_y (phase encode) is stepped to a new level before the next TR.
- 4 Repeat until all phase encoding steps are collected and k-space is filled.

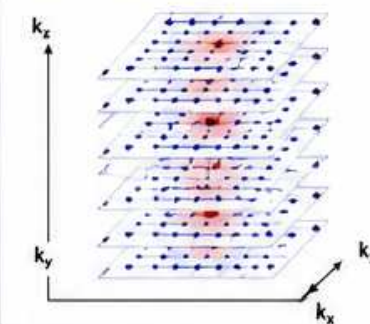


HOW PHASE ENCODING STEPS FILL K-SPACE



3D K-SPACE (ADDS SLICE ENCODING)

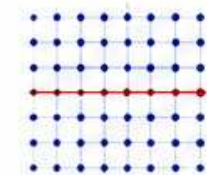
A third gradient (G_z) is used to select a slice and step through k-space planes (or partitions).



- F** If each slice (k_z step):
1. Select slice with G_z
 2. Fill 2D k-space using G_y (phase encode) and G_x (readout)
 3. Move to next slice (next G_z step)
- One k-space plane (2D)
 - Multiple k_z steps (slices)

WHERE AM I IN K-SPACE?

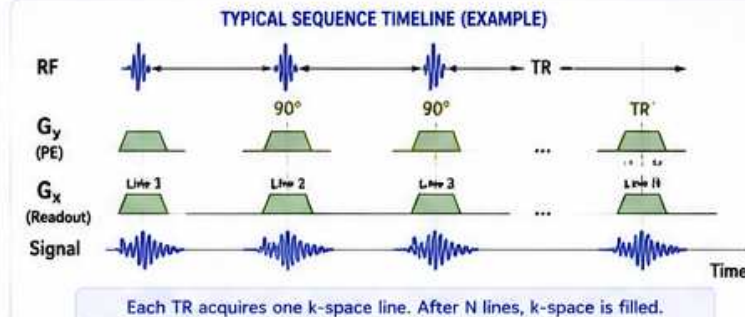
During acquisition, the system knows exactly which line of k-space is being sampled.



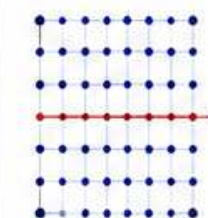
Current k-space line =
(k_x samples along readout,
 k_y phase step, k_z slice)

KEY POINTS

- ✓ Readout (G_x) fills k-space from $-k_{x,max}$ to $+k_{x,max}$ in one line.
- ✓ Phase encoding (G_y) steps between TRs to move to the next line.
- ✓ Center of k-space (low frequencies) is usually acquired in the middle of the scan to capture the highest signal when T_2^* decay has had less time to reduce it.
- ✓ More phase encoding steps (higher matrix) = more lines = better resolution but longer scan time.



CENTER LINE MATTERS



The center line ($k_y = 0$) contains the most important low spatial frequency data (overall signal & contrast).

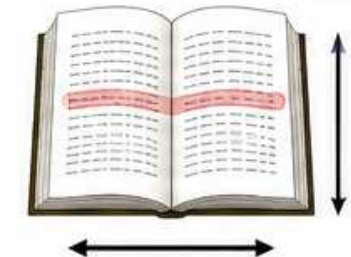
If the center is missed or discarded by artifact, contrast and image quality suffer significantly.

REAL-WORLD ANALOGY

Think of reading a book:
 G_x reads the letters across a line (left to right).

G_y moves you to the next line up or down.

After many lines, the whole page (image) is complete.



KEY TAKEAWAY

K-space is filled line by line: G_x samples one line (frequency encoding) while G_y steps between lines (phase encoding). After many TRs and phase steps (and k_z steps for 3D), k-space is complete and ready for Fourier Transform.

WHAT'S NEXT? ➔

Next: Section 1.3 – Center vs. Outer K-Space
We'll see how different parts of k-space control contrast and detail in the image.

